SRIVARMA BATTINI

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About Me

Motivated Computer Science and Engineering (AI) student with a strong foundation in mathematics, analytical thinking, and problem-solving. Passionate about Artificial Intelligence and driven to continuously learn emerging technologies and concepts. Enjoys applying theoretical knowledge to solve real-world problems through practical AI solutions. Eager to develop innovative solutions, and grow as an AI Engineer.

Education

Amrita Vishwa Vidyapeetham **Aug 2023 – Present**
B.Tech in CSE (Artificial Intelligence) GPA: 8.91

Sri Chaitanya Junior College **2021 – 2023**
Higher Secondary Education Percentage: 97.4%

Technical Skills

Programming Languages: Python, SQL
Machine Learning: Regression, Classification, Scikit-learn, XGBoost, Feature Engineering
Deep Learning: PyTorch, CNNs, RNNs, LSTMs, U-Net, Transformers
Natural Language Processing: BERT, DeBERTa, SBERT, Text Classification, Text Embeddings
Reinforcement Learning: Q-Learning, RMAB, Whittle Index, LPQL
Generative AI & LLMs: RAG, Llama 3.1, Prompt Engineering, Groq API, ChromaDB
Databases: PostgreSQL, MongoDB, SQLite
Cloud & Development: Docker, Kubernetes, FastAPI, REST APIs, Git

Projects

SmartRecall: RL-Based Spaced Repetition Scheduling **Aug 2026 – Nov 2026**
Skills: Python, Reinforcement Learning, Q-Learning, RMAB, Whittle Index, LPQL

- Formulated memory retention scheduling as a **Restless Multi-Armed Bandit (RMAB)** problem using recall probability and memory half-life modelling.
- Explored **Whittle Index-based scheduling** and Q-learning approaches to prioritize memory reviews under a limited daily review budget.
- Developed a simulation environment incorporating memory state transitions, reward computation, and evaluation against spaced repetition baselines.

EmailIQ: Controlled Email Tone and Politeness Optimization **Mar 2026 – May 2026**
Tech: DeBERTa, RAG, SBERT, ChromaDB, Llama 3.1, Groq API, Python

- Built an NLP system using **7 fine-tuned DeBERTa models** to classify tone, intent, politeness, sarcasm, passive aggression, urgency, and power dynamics in workplace emails.
- Designed a **RAG pipeline** with SBERT embeddings and ChromaDB, integrating Llama 3.1 via Groq API to generate professional rewrites while preserving original intent.
- Engineered risk assessment and professionalism scoring modules for actionable workplace communication feedback.

Constraint-Based Intelligent Travel Planning System **Oct 2025 – Jan 2026**
Tech: React, FastAPI, REST API, MongoDB, Python, Dynamic Programming, TSP

- Built a full-stack application (React + FastAPI + MongoDB) generating personalized itineraries based on user constraints such as budget, duration, and interests.
- Implemented **Dynamic Programming** for K-best itinerary generation, **0/1 Knapsack** for place selection, and **TSP** (Nearest Neighbour + 2-opt) for route optimization.

Competitive Programming Repository

- Maintained a structured repository of DSA and competitive programming solutions in Python covering graphs, dynamic programming, trees, and sorting with complexity analysis.

Certificates

- Google Cloud Computing Foundations Certificate – Google Cloud
- Elements of AI – MinnaLearn & University of Helsinki
- Deep Learning with PyTorch – DataCamp